FOR A COUNTABLE MODULE OVER A COUNTABLE RING, FINITE PRESENTATION IS EQUIVALENT TO THE CANONICAL MAP M TENSOR R^N -> M^N BEING AN ISOMORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The kernel-countability step is proved by 'arguing as in the proof of Algebra, Proposition algebra-proposition-fp-tensor' - the same enumeration argument as the finite-generation case applied to K; recorded as mechanical analogy, content verbatim.

Editorial note. The label lemma-countable-fp-tensor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The canonical map M \otimes_R R^N -> M^N and its finitely generated counterpart (Algebra, proposition-fp-tensor); countable modules and countable rings
- Same-chapter lemma-countable-fg-tensor (L48287 cluster): for countable M, finite generation iff the map is surjective

Source reference. Original proposition 059K: sources/stacks-chow/tags.txt, source lines 6818–6818. Prerequisite pointer invoked at 988; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AL9: sources/stacks-examples/pinned-original-source.tex, source lines 963–963. Prerequisite pointer invoked at 989; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a countable ring. Let M be a countable R-module. Then M is finitely presented if and only if the canonical map $M \otimes_R R^{\mathbf{N}} \to M^{\mathbf{N}}$ is an isomorphism.*

Proof. If M is a finitely presented module, then the map is an isomorphism by Algebra, Proposition 059K. Conversely, assume the map is an isomorphism. By Lemma 0AL9 the module M is finite. Choose a surjection $R^{\oplus m} \to M$ with kernel K. Then K is countable as a submodule of $R^{\oplus m}$. Arguing as in the proof of Algebra, Proposition 059K we see that $K \otimes_R R^{\mathbf{N}} \to K^{\mathbf{N}}$ is surjective. Hence we conclude that K is a finite R-module by Lemma 0AL9. Thus M is finitely presented. □

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